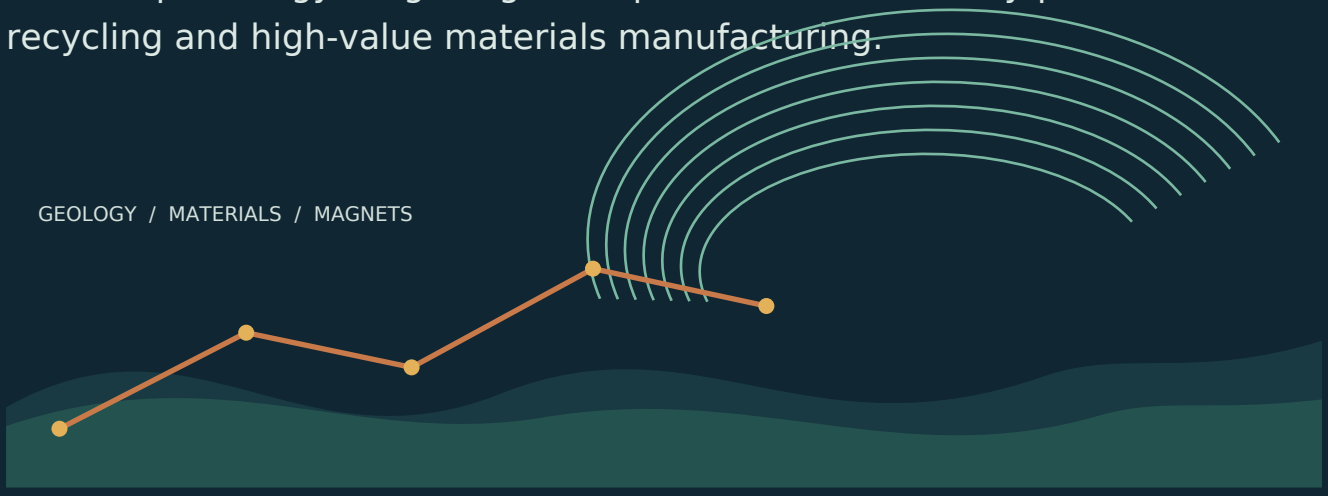
A BRITISH CRITICAL MATERIALS OPPORTUNITY

Building a British Rare Earth Supply Chain

A start-up strategy for geological exploration, industrial by-products, recycling and high-value materials manufacturing.



INVESTMENT THESIS

Start with the strongest evidence. Build value beyond the mine.

Prepared as a strategic briefing for a prospective British rare-earth minerals venture. The company name, ownership structure and funding assumptions have not been supplied; no invented entity details or financial forecasts are included.

01 / EXECUTIVE BRIEF

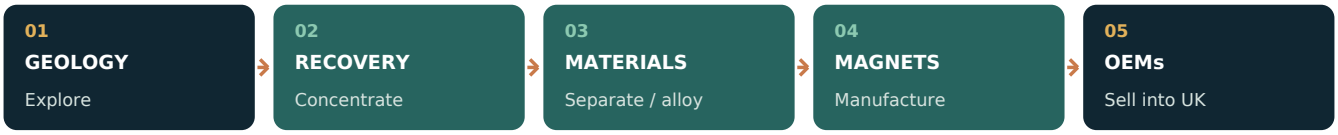
The best start-up is a supply-chain company first, a mining company second

Britain has documented rare-earth enrichment in north-west Scotland and research-backed opportunities to investigate rare earths in Cornish mineral-processing residues. Yet no commercially economic UK rare-earth reserve is established by the evidence reviewed here. The investable proposition is therefore staged: test domestic feedstocks while building processing, recovery and customer relationships that can work with recycled and responsibly sourced international material. [1-5]

Investment thesis

Prioritise neodymium-praseodymium (NdPr) and permanent-magnet feedstocks; make Loch Loyal the lead primary exploration target; test Cornish residues for by-product recovery; and develop partnerships around magnet recycling and downstream alloy / component demand.

What the venture should be built to do



Value is captured at several stages - not only at the mine gate.

Four operating principles

PRINCIPLE	WHAT IT MEANS FOR THE START-UP
Evidence before acreage	Do not treat a mineral occurrence, high-grade sample or geological analogy as a resource. Buy data, sample systematically and drill only when the target model is testable.
Recover value in stages	Evaluate secondary feedstocks, separation, metal / alloy routes and offtake alongside mining. The company should not depend on a successful greenfield mine to create its first revenue.
Sell to a defined customer	Anchor product specifications and recovery flowsheets to magnet, alloy, engineering and recycling customers before scaling capital expenditure.
Design for responsible operations	Screen thorium-bearing minerals, reagents, effluents, tailings, water, biodiversity and planning constraints at the earliest stage.

Key policy context. The UK Government Vision 2035 strategy aims for 10% of annual aggregate critical-mineral demand to be met by domestic production, 20% through recycling, and no more than 60% of aggregate demand to be supplied by any one country by 2035. These are cross-mineral ambitions, not specific rare-earth quotas. [1]

02 / PRODUCT STRATEGY

Target the magnet chain, not the periodic table

For a British start-up, the strongest demand logic lies in rare earths that enable high-performance permanent magnets. The commercial target is not simply the highest spot-price element: it is the combination of end-use demand, supply vulnerability, recoverability, product specification and the ability to secure an offtake buyer.

PRIORITY	MATERIAL	WHY IT MATTERS	START-UP POSITION
1	**Neodymium + praseodymium (NdPr)**	Core rare-earth inputs for NdFeB permanent magnets used in EV motors, generators, robotics and industrial systems.	Primary commercial lens: assay and model NdPr recoverability; align specifications with alloy and magnet customers.
2	**Dysprosium + terbium (Dy / Tb)**	Can improve high-temperature performance in specialised NdFeB magnets; strategically relevant but not necessarily abundant in light-REE deposits.	Track carefully in assays and recycled feedstocks. Do not assume known UK prospects contain economic Dy / Tb.
3	**Samarium (Sm)**	Important for specialist samarium-cobalt magnets where heat tolerance and reliability matter.	Niche materials / defence and aerospace supply partnerships; not a stand-alone mining thesis at this stage.
4	**Cerium + lanthanum (Ce / La)**	Industrial uses include catalysts, polishing, glass and chemicals; often important co-products in light-REE systems.	Model as co-products and test marketable specifications; do not overstate value per tonne.
5	**Yttrium, scandium and other REEs**	Potential specialist uses in ceramics, alloys and advanced materials. Market depth and price vary materially.	Measure in the full analytical suite; treat as optionality until a customer and recovery route exist.

Product decisions the founder should make early

- Feedstock: mineral concentrate, kaolin-processing residue, end-of-life magnets, alloy scrap, or a blend? Each requires a different process and permitting case.
- Product: mixed rare-earth carbonate / oxide, separated oxide, metal, master alloy, magnet powder or a recycling service? Higher-value stages usually require tighter specifications, more capital and more process knowledge.
- Customer: magnet makers, alloy producers, automotive supply chains, wind, industrial motor makers, defence primes or a materials trader? Obtain technical conversations before defining the plant.

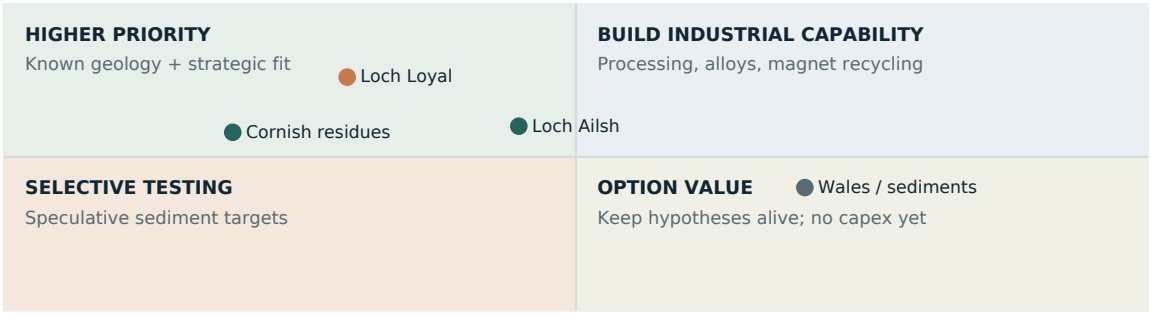
Commercial discipline

Do not multiply total rare-earth oxide (TREO) grade by a single headline rare-earth price. Model every recoverable element, payability, process recovery, impurity penalty, reagent and energy cost, and the cost of managing waste.

03 / UK GEOLOGY

Rank targets by evidence - and keep the speculation labelled

The UK Critical Minerals Intelligence Centre has published a national-scale prospectivity assessment intended to guide detailed research and exploration. The evidence supports targeted investigation, not a claim that the UK already has an economic rare-earth resource. [2]



Qualitative screening view only - not a measured grade, resource estimate or investment rating.

TARGET	EVIDENCE BASE	PRIORITY	MAIN QUESTION
Loch Loyal / Cnoc nan Cuilean, Sutherland	Peer-reviewed work identifies REE-bearing allanite and locally enriched biotite-magnetite veins; reported bulk-rock samples reached up to 2% total rare-earth oxides in selected veins. [3]	**Very high for primary exploration**	Are enriched veins continuous, mappable and recoverable at useful scale?
Loch Ailsh, north-west Highlands	Historical BGS-related studies identify elevated La + Ce in syenitic rocks. Results are point samples, not a reserve estimate. [3]	**High**	Is there sufficient grade, tonnage and metallurgical recovery?
Loch Borralan, north-west Highlands	REE-bearing apatite and other accessory minerals reported; existing reviews describe variable enrichment. [3]	**Medium / selective**	Can a distinct zone outperform the known limitations?
Cornwall - kaolin residues / granite systems	Research on a kaolin plant identified a micaceous residue with around 170 ppm light rare earth elements (La, Ce, Nd); monazite and other metal-bearing minerals were observed. [4-5]	**High for low-cost testing**	Can recovery from a consistent residue stream produce a saleable product?
Central Wales / sediments	Monazite occurrences are recorded, but no commercial-scale deposit is established by sources reviewed here.	**Low-cost screening only**	Do any local concentrations justify follow-up beyond reconnaissance?

Important: The matrix is a qualitative exploration-screening device. It is not a formal mineral-resource estimate, economic rating or investment recommendation. Reported high sample grades are not representative grades unless sampling demonstrates that they are.

04 / LEAD EXPLORATION TARGET

Loch Loyal: the strongest first drill-question

At Cnoc nan Cuilean in the Loch Loyal Syenite Complex, researchers documented rare-earth enrichment associated with allanite, including biotite-magnetite veins. The highest reported bulk-rock values reached up to 2% total rare-earth oxides in those selected veins. Their lateral extent is uncertain because the veins are poorly exposed. [3]

> **What makes it interesting**
>
> A clear, testable geological model: light rare-earth elements concentrated in allanite-bearing rocks and locally enriched hydrothermal veins. The next value-creating step is not a mine plan; it is establishing geometry, continuity, full element distribution and mineral-processing response.

Proposed technical work package

WORK PACKAGE	METHOD	DECISION IT UNLOCKS
1. Data consolidation	Compile historic maps, assays, cores, academic work and 3D models; standardise sample coordinates and analytical units.	Where are the known high-value zones and data gaps?
2. Geological mapping	Map intrusive lithologies, vein orientations, alteration, exposure and structural controls.	Can a coherent mineral-system model be defined?
3. Multi-element sampling	Representative channels / outcrop samples; full REE suite plus U, Th, P, Fe and relevant impurity suite with QA/QC.	What is the grade distribution and element mix beyond headline samples?
4. Mineralogy + metallurgy	Mineralogical liberation, grain size, concentration tests and early hydrometallurgical testwork; screen thorium-bearing phases.	Can valuable REEs be separated at an acceptable cost and waste profile?
5. Geophysics + drilling	Only after mapping and sampling support targets; select a small, hypothesis-led drilling programme.	Is there repeatable three-dimensional continuity?

The crucial caveats

- Known enrichment is dominated by light REEs in allanite-bearing assemblages. It is not evidence that the target is rich in dysprosium or terbium.
- The vein observations are locally high-grade but their lateral extent is poorly constrained. One or a few enriched samples cannot establish a mineable tonnage.
- Allanite, apatite and associated minerals require real mineral-processing tests. The presence of thorium-bearing phases could affect permitting, residue handling and costs.

Primary technical source: Hughes et al., “Enrichment of Rare Earth Elements during magmatic and post-magmatic processes: a case study from the Loch Loyal Syenite Complex, northern Scotland,” Contributions to Mineralogy and Petrology (2013). [3]

05 / SECONDARY FEEDSTOCKS

Cornwall may offer a faster test than a new mine

A 2015 study of output streams from a Cornish kaolin plant found that rare metals and light rare-earth elements were pre-concentrated in a micaceous residue. The residue contained around 170 ppm La, Ce and Nd combined as described by the paper, and monazite was identified among the relevant metal-bearing minerals. This is an investigation lead - not proof of a commercial by-product reserve. [4]

Why test residues first?

- Potentially avoid some of the costs of mining fresh rock, since material has already been excavated and processed.
- Work with operators to obtain representative samples, mass-flow data, historical tailings records and long-term supply agreements.
- Run bench-scale gravity / magnetic / flotation and downstream extraction tests against an explicit product specification.
- Measure U and Th, reagent consumption, water use, residues and tailings stability; cost the full environmental management plan.

> **First commercial experiment**
> Secure permission to sample one defined residue stream, build a mass balance, run an independent assay and concentration test, then obtain indicative product specifications from potential buyers. Do not build a plant until the feedstock contract and recovery economics are proven.

Granite systems and other hypotheses

Research on the Carnmenellis pluton shows that monazite hosts much of the light rare-earth inventory in Cornish biotite granite, while xenotime, apatite and zircon host important portions of the heavy rare-earth inventory. This establishes geological occurrence, not economic concentration. [5]

Central Wales and alluvial / coastal sediments should be kept as lower-priority reconnaissance hypotheses. Resistant heavy minerals such as monazite can be concentrated by sedimentary processes in some geological settings; however, the sources reviewed here do not establish a commercially exploitable British rare-earth mineral-sands deposit. Any work should begin with desk study, non-invasive screening and environmental constraints.

Feedstock ranking for a start-up

FEEDSTOCK	UPSIDE	PRINCIPAL RISK	NEXT ACTION
Kaolin residues / mine wastes	Potentially lower mining burden; existing stream may be accessible.	Low concentration, variability, rights / ownership, residue tonnage, radioactivity.	Sample one stream; establish mass balance and supply rights.
End-of-life magnets / manufacturing scrap	Known product stream; direct link to circular supply chains.	Collection, sorting, contamination, magnet availability and competing recyclers.	Partner with dismantlers, OEMs and recycling projects.
Primary Scottish mineralisation	Potential domestic mine supply and strategic differentiation.	Resource continuity, metallurgy, permitting, capex and time.	Stage-gated exploration; no premature mine capex.

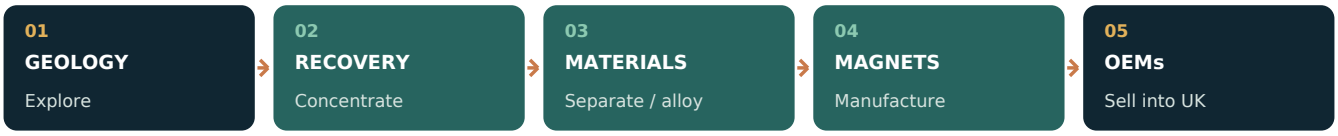
06 / START-UP BUSINESS MODEL

Three connected routes to revenue

A resilient venture can build customer relationships and technical capability before domestic mining is proven. The right sequence depends on founder expertise, access to feedstock, capital availability and buyer specifications.

ROUTE	REVENUE MODEL	CAPABILITIES NEEDED	ROLE IN PORTFOLIO
A. Feedstock qualification + recovery	Paid assay / sorting services, toll processing, recovery fees or sale of a specification-controlled concentrate.	Sampling QA/QC, mineralogy, mass balance, process testwork, waste management and customer audits.	Best candidate for low-cost pilots where a defined stream and partner exist.
B. Magnet circularity	Collection and dismantling contracts, recovery service, recycled alloy / powder sales, offtake partnerships.	Reverse logistics, product identification, safe dismantling, metal recovery and traceability.	Builds a secondary feedstock and links to UK industrial initiatives.
C. Domestic exploration	Longer-term sale of concentrate / oxide or strategic value in a discovered resource, subject to permits and feasibility.	Exploration management, resource estimation, metallurgy, land access, environmental planning and mine finance.	Potential high strategic value; longest lead time and highest uncertainty.

Where to compete in the chain



Value is captured at several stages - not only at the mine gate.

Britain already has relevant industrial capability. Less Common Metals has reported commercial production of neodymium and NdPr metal in the UK and commercial NdFeB alloy activity, while the company was acquired by USA Rare Earth in 2025. This demonstrates that key midstream know-how exists in the UK, but also means a start-up must define a clear niche rather than assume an empty market. [6]

The University of Birmingham opened a rare-earth magnet separation and recycling facility in the West Midlands in January 2026, and the university announced the £6.5 million, three-year REACT-UK programme in April 2026 to develop a circular automotive magnet supply chain. A start-up should explore collaboration, feedstock supply and complementary services instead of duplicating research infrastructure without a customer case. [7-8]

Suggested positioning

“A British rare-earth feedstock and materials company, built around recoverable domestic residues, recycled magnets and selective exploration - supplying traceable, specification-led materials to UK industrial customers.”

07 / EXECUTION PLAN

A stage-gated 24-month roadmap

The purpose of the first two years is to convert geological and industrial hypotheses into investable, auditable evidence. Capital should be released against technical and commercial milestones, not calendar time alone.

PHASE	TIMING	CORE ACTIONS	GATE TO PROCEED
0. Define the company	Months 0-2	Choose a primary lane (residues, recycling or exploration); hire / contract a mineral-processing lead; map relevant land, data and customer rights.	One clear product thesis, a target customer set and a credible operating budget.
1. Access and data	Months 1-4	Obtain historic datasets and sample access; interview potential buyers; establish QA/QC standards and chain-of-custody.	Access agreements and representative samples; no major fixed assets.
2. Bench-scale proof	Months 3-9	Independent assays, mineralogy, concentration / leach testing, mass balance, impurity and waste screening.	Repeatable recovery data and indicative buyer interest.
3. Commercial design	Months 6-12	Process flowsheet, cost estimates, permitting review, feedstock agreements, product specs, preliminary offtake discussions.	Credible gross-margin model under downside assumptions.
4. Pilot / field work	Months 9-18	Pilot processing or targeted mapping / drilling depending on chosen route; lifecycle and residue-management analysis.	Scale-up decision with independent technical review.
5. Finance the next stage	Months 18-24	Secure strategic partner, grants or project finance; finalize plant / exploration design and contracts.	Invest only when supply, process, customer and environmental cases align.

Indicative diligence budget buckets

BUDGET BUCKET	WHAT IT COVERS	RULE OF THUMB
Data, land and samples	Data licensing, access, sample logistics, landowner / operator agreements.	Spend early to reduce uncertainty; avoid speculative land accumulation.
Independent lab and metallurgy	Assays, standards / blanks / duplicates, mineralogy, bench tests, process mass balance.	Protect QA/QC; repeat results matter more than a headline assay.
Commercial and regulatory	Buyer specifications, offtake, waste / radiological screening, permitting, environmental baseline.	Run in parallel with technical work, not after it.
Pilot / drilling	Small-scale pilot equipment or carefully targeted drilling.	Only after prior gates; avoid premature plant commitment.

No budget total is stated because project route, scale, site, feedstock access and financing assumptions are not yet defined. A reliable budget requires quotes and site-specific scope.

08 / RISK & GOVERNANCE

The risks that can destroy the thesis

The company should make uncertainty visible to investors and partners. Geological potential is only one risk; processing, customer qualification, feedstock rights, environmental liabilities and market concentration can be equally decisive.

RISK	WHY IT MATTERS	MITIGATION / EVIDENCE REQUIRED
Geology / tonnage	High-grade samples may be local and unrepresentative; continuity may be absent.	QA/QC sampling, geological model, independent technical review, staged drilling and competent-person reporting if a resource is later declared.
Metallurgy / recovery	Rare earth minerals may be difficult to liberate, separate or refine; recoveries drive economics.	Identify host minerals early; run variability tests; develop a full mass balance and product specification.
Co-product assumptions	La / Ce abundance does not mean premium-value revenue; element prices and markets differ.	Build element-by-element revenue; stress test payability, by-product credits and price volatility.
Radioactivity / residues	Monazite, allanite and associated assemblages may contain Th or U; waste streams can create regulatory costs.	Assay U / Th early; engage environmental and radiological specialists; design a compliant residue route.
Environmental / planning	Scottish Highland and Cornish sites may face sensitive habitats, landscape, water and community constraints.	Screen designation, land rights, water balance, tailings, biodiversity and stakeholder consent before committing to a site.
Feedstock security	Residues and scrap are not automatically available, consistent or legally owned by the start-up.	Secure contracts, volumes, sampling rights, quality windows and chain of custody.
Customer qualification	Magnet / alloy users require repeatable purity, traceability and technical performance.	Co-design specifications with buyers; use qualification samples and letters of intent before scale-up.
Competition / geopolitics	Incumbent suppliers may have scale and policy support; prices can compress quickly.	Compete on resilience, traceability, specialised grade, circularity and contracted service, not only commodity price.

First-year KPI scorecard

TECHNICAL
Representative samples tested; assay QA/QC passed; mineral host identified; recovery and mass balance established; process variability measured.

Investment committee rule

No mine, pilot plant or full-scale separation plant should be sanctioned solely on the basis of geological occurrence or headline assay. Require independent validation of feedstock, recovery, product quality, customer demand and regulatory route.

09 / RECOMMENDATION

Build optionality before building a mine

The best initial strategy for a British rare-earth start-up is to develop capabilities that can earn revenue and validate material flows even if domestic mining takes longer than expected. The recommended combination is: selective exploration at Loch Loyal, a low-cost residue assessment in Cornwall, and partnerships in recycled magnet feedstock and downstream materials.

Recommended priorities for the founding team

ORDER	DECISION	REASON
1	Run a structured data room and define one primary operating lane.	Avoid spreading early capital across mining, refining and recycling simultaneously.
2	Secure a representative feedstock stream and independent lab programme.	Fastest route to useful evidence about grade, variability and recoverability.
3	Interview at least 10 potential buyers / users and collect specifications.	Product design should be driven by what a real customer can qualify and purchase.
4	Keep Loch Loyal on a disciplined, staged exploration track.	Best documented primary rare-earth target in the sources reviewed, but resource scale is uncertain.
5	Prepare a downside-first technical-economic model.	Stress-test recovery, payability, electricity, reagents, waste management, capex and commodity prices.

Bottom line

Britain does not need to prove a giant mine before it can build a useful rare-earth business. It needs evidence-led exploration, reliable secondary feedstock, process know-how, industrial customers and robust environmental management. A mine would be a major upside if the geology and economics prove out - not the only route to a viable company.

Scope and limitations

This document is a strategic starting point, not a bankable feasibility study, mineral resource estimate, legal or environmental opinion, investment solicitation, or forecast of commodity prices. Geological observations cited are reported occurrences and selected samples, not proved reserves. All current company / policy information and source links were checked for this briefing on 13 October 2026. Before external circulation, the founding team should add its legal entity name, management credentials, site-specific data, financial assumptions and independent technical review.

10 / SOURCES & RESEARCH BASIS

Evidence behind the strategy

The conclusions above combine official UK critical-minerals policy, the UK Critical Minerals Intelligence Centre prospectivity assessment, peer-reviewed geological studies, and current UK industrial / research activity. Sources support the specific claims noted in the text; the exploration rankings and start-up roadmap are strategic judgements, not published resource classifications.

How to read the evidence

Selected geological assays and mineral occurrences are useful for generating targets, not declaring reserves. The start-up must commission representative sampling, mineralogical work, metallurgical tests and independent technical review before making any resource or economic claims.

SELECTED SOURCES

[1] UK Department for Business and Trade (2025; updated 23 Jan 2026), Vision 2035: Critical Minerals Strategy. Policy targets for domestic production, recycling and supplier diversification.

[2] UK Critical Minerals Intelligence Centre (2023), Potential for Critical Raw Material Prospectivity in the UK. National-scale prospectivity assessment.

[3] Enrichment of Rare Earth Elements during magmatic and post-magmatic processes: a case study from the Loch Loyal Syenite Complex, northern Scotland, Contributions to Mineralogy and Petrology (2013).

[4] Dehaine, Q. & Filippov, L.O. (2015), Rare earth (La, Ce, Nd) and rare metals as by-product of kaolin production, Cornwall: Part 1, Minerals Engineering 76, 141-153.

[5] Jefferies, N.L. (1985), The distribution of the rare earth elements within the Carnmenellis pluton, Cornwall, Mineralogical Magazine 49, 495-504.

[6] Less Common Metals, Metal and alloy production in the Western World: A supply chain alternative; company history notes acquisition by USA Rare Earth in 2025.

[7] University of Birmingham (15 Jan 2026), World-leading rare earth magnet recycling facility launches in West Midlands.

[8] University of Birmingham (24 Apr 2026), Fully circular UK supply chain for rare earth magnets with high grade automotive applications (REACT-UK).